So far, we have provisioned VPCs, subnets, and EC2 instances in the AWS cloud, and discussed their security hardening. What about the network connections between the public cloud and the on-premises data centers? Let's look at that in the next section.

Understanding Amazon Direct Connect

Amazon Direct Connect is a network service that provides a dedicated and private connection between your on-premises data center environment and the virtual data centers in the cloud. This connection is a private, high-speed, low-latency connection that is not accessible over the public internet. With Amazon Direct Connect, the connection between your on-premises environment and the AWS cloud will have guaranteed bandwidth and performance.

To set up AWS Direct Connect, you'll need to follow these steps:

1. **Review the prerequisites**: Before you begin, make sure you have an AWS account and access to the AWS Management Console, a Direct Connect location, and a device that can connect to the Direct Connect location.
2. **Request a connection**: Log in to the AWS Management Console and go to the Direct Connect console. Choose a Direct Connect location and request a new connection. Provide details such as your connection name, port speed, and virtual private gateway.
3. **Choose a Partner**: Choose a Direct Connect Partner who can provide a network connection between your on-premises environment and the Direct Connect location.
4. **Configure the connection**: Once the connection has been approved by the Direct Connect Partner, you'll receive a **Letter of Authorization and Connecting Facility Assignment** (**LOA-CFA**) document. You'll need to provide this document to the colocation facility where your equipment will be located. You'll also need to configure your routers to connect to the Direct Connect location.
5. **Test the connection**: After the connection is set up, test it to make sure it's working properly. You can do this by creating a test VPC and connecting to it from your on-premises environment.
6. **Start using Direct Connect**: Once you're confident that the connection is working as expected, you can start using Direct Connect to connect to your AWS resources.

More details are available at `https://docs.aws.amazon.com/directconnect/latest/UserGuide/getting_started.html`.

To secure the traffic over Amazon Direct Connect, you can set up a VPN over an Amazon direct connection, extend your private network into AWS, and securely access resources running in the cloud. To set up a VPN over Amazon Direct Connect, you will need to configure a virtual private gateway in your AWS VPC and a customer gateway in your on-premises environment and configure a VPN connection between the two gateways. More details are available at `https://aws.amazon.com/premiumsupport/knowledge-center/create-vpn-direct-connect/`.

As a basic network service, DNS plays a critical role in networking and the internet. When we extend our network from on-premises to the cloud, how does DNS work in the cloud? We will examine this in the next section.

Understanding Amazon DNS – Route 53

Amazon Route 53 is a scalable and highly available DNS web service provided by AWS. Some details about AWS Route 53 are as follows:

- It provides translation from human-readable names (such as `example.com`) into IP addresses that computers use to connect to network resources
- It can also be used to route traffic to AWS resources, such as EC2 instances, S3 buckets, and load balancers
- It offers features such as latency-based routing, geo DNS, health checks, and DNS failover, which help improve the performance, availability, and reliability of applications
- It integrates with other AWS services, such as CloudWatch and CloudTrail, to provide additional monitoring and logging capabilities

Amazon Route 53 supports the following routing policies:

- **Simple routing**: Uses one Route 53 server
- **Weighted round-robin routing**: Assign weights to Route 53 server rotations
- **Latency routing**: Route traffic based on latencies
- **Geolocation routing**: Route traffic based on the location of users
- **Geoproximity routing**: Route traffic based on the location of resources
- **Failover routing**: Route traffic to the failover site when the primary site is not reachable
- **Multivalue answer routing**: Distributes DNS responses across multiple IP addresses – up to eight healthy records can be selected at random

More details on Amazon Route 53 can be found at `https://aws.amazon.com/route53/`.

In the next section, we will cover the AWS CDN, which provides efficient delivery of cloud content, such as photos and PDF files.

Understanding the Amazon CDN

The Amazon CDN is called Amazon **CloudFront** (**CF**). It is a global delivery service that leverages edge locations to securely deliver content with low latency and high speed. CF is integrated with AWS cloud services, and has the following features:

- It is a global network of edge locations that cache and deliver content (such as web pages, videos, and software downloads) to end users with low latency and high data transfer speeds.
- It can be used to distribute both static and dynamic content from AWS and non-AWS origin servers, such as EC2 instances, S3 buckets, and on-premises servers.
- It offers features such as content compression, SSL/TLS encryption, and custom domain names, which help improve the security, reliability, and performance of applications.
- It has a pay-as-you-go pricing model, with no upfront costs or minimum fees. The cost is based on the amount of data transferred, the number of requests made, and the edge locations used.

More details about Amazon CF can be found at `https://aws.amazon.com/cloudfront/`.

Summary

In this chapter, we discussed the Amazon cloud network – VPC, Direct Connect, Route 53, and the CDN, CF. We focused on building a small AWS cloud, by provisioning VPCs, subnets, EC2 instances, and peering VPCs, and setting up internet access for the EC2 instances. We discussed AWS VPC network security and compared the differences between SGs and NACLs. At the end of the chapter, we briefly looked at AWS Direct Connect, providing a connection between on-premises and the Amazon cloud, Amazon Route 53, and Amazon CF concepts and features. In the next chapter, we will discuss database concepts and introduce Amazon cloud database services.

Practice questions

1. What is the main function of an **Internet Gateway** (**IGW**) in AWS VPC?

    A. To provide **Network Address Translation** (**NAT**) for instances in the VPC

    B. To allow instances in the VPC to communicate with other resources outside the VPC

    C. To restrict access to the VPC to only authorized users or resources

    D. To provide load-balancing capabilities for instances in the VPC

2. Which of the following is not a valid use case for a NAT gateway in Amazon VPC?

    A. Allowing instances in a private subnet to access the internet

    B. Enabling communication between two VPCs in different regions

    C. Enabling communication between a VPC and an on-premises network

D. Enabling communication between a VPC and a third-party service that requires a whitelisted IP address

3. Which of the following statements is true about AWS security groups?

A. Security groups are stateless, meaning that inbound and outbound rules must be specified separately

B. Security groups are attached to individual instances, not to a VPC

C. Security groups can be used to restrict inbound and outbound traffic based on IP addresses or port numbers

D. Security groups can be used to allow traffic from a specific IP address range requiring a VPN connection

4. Which of the following is a key difference between an AWS SG and NACL?

A. Security groups can only be applied at the subnet level, while NACLs can be applied at the instance level

B. Security groups are stateful, meaning that they automatically allow return traffic for allowed inbound traffic, while NACLs are stateless

C. Security groups can only allow traffic based on IP addresses, while NACLs can allow or deny traffic based on IP addresses, port numbers, or protocols

D. Security groups can be used to create complex security policies that can filter traffic at multiple layers, while NACLs are simpler and more restrictive in nature

5. Which statement about AWS Direct Connect is true?

A. It is a service that allows you to securely connect your on-premises network to AWS over the public internet

B. It is a service that allows you to create a private, dedicated network connection between your on-premises infrastructure and AWS

C. It is a service that enables you to register domain names and route traffic to AWS resources

D. It is a service that provides managed DNS hosting and traffic management capabilities for your domain names

6. Which of the following is not a valid reason to use AWS Direct Connect?

A. To reduce bandwidth costs for accessing AWS services

B. To improve the reliability and performance of your network connections to AWS

C. To improve the security of your network connections to AWS

D. To access AWS services from a location where internet access is not available

7. What is AWS Route 53?

   A. A service that enables you to register domain names and route traffic to AWS resources

   B. A service that provides managed DNS hosting and traffic management capabilities for your domain names

   C. A service that enables you to establish a private, dedicated network connection between your on-premises infrastructure and AWS

   D. A service that allows you to securely connect your on-premises network to AWS over the public internet

8. Which of the following is a valid use case for Route 53 traffic routing policies?

   A. Distributing traffic evenly across multiple EC2 instances in a single Availability Zone

   B. Directing traffic to the nearest AWS region based on the location of the user

   C. Balancing traffic across multiple AWS accounts using weighted routing

   D. Limiting the amount of traffic that can be sent to a specific AWS resource

9. What is AWS CloudFront?

   A. A service that allows you to run containers on AWS using Docker

   B. A service that enables you to manage and store large amounts of data in the cloud

   C. A **Content Delivery Network** (**CDN**) that speeds up the delivery of your static and dynamic web content

   D. A service that provides serverless computing capabilities for running code in response to events

10. Which of the following is a valid use case for AWS CloudFront?

    A. Storing and managing large volumes of data for analysis using AWS analytics services

    B. Running containers on AWS using Kubernetes

    C. Hosting a website that requires complex server-side processing and database access

    D. Accelerating the delivery of content to users worldwide, especially those in remote or high-latency locations

Answers to the practice questions

1. B
2. B
3. C
4. B
5. B

6. D
7. B
8. B
9. C
10. D

Further reading

For further insights into what you've learned in this chapter, refer to the following links:

- `https://aws.amazon.com/vpc`
- `https://docs.aws.amazon.com/vpc/index.html`
- `https://aws.amazon.com/vpc/faqs/`
- `https://aws.amazon.com/getting-started/hands-on/getting-started-create-vpc/`
- `https://docs.aws.amazon.com/directconnect/index.html`
- `https://aws.amazon.com/directconnect/faqs/`
- `https://aws.amazon.com/getting-started/hands-on/getting-started-dc/`
- `https://docs.aws.amazon.com/cloudfront/index.html`
- `https://aws.amazon.com/cloudfront/faqs/`
- `https://aws.amazon.com/getting-started/hands-on/get-started-with-cloudfront/`
- `https://docs.aws.amazon.com/Route53/index.html`
- `https://aws.amazon.com/route53/faqs/`
- `https://aws.amazon.com/getting-started/hands-on/get-started-route-53/`